BIL216 — Signals and Systems

Final Project Phase 2 Report

Emo Challenge 2026 — Speech Emotion Classification

Course	BIL216 — Signals and Systems
Semester	2025–2026 Spring
Group	9
Submission	Phase 2 — Research & Development

Group Members

Name	Student ID	Role
Ahmet Akin	230611038	Feature Engineering & Software Implementation
Berivan Demir	230611030	Model Design, Ensemble & Hyperparameter Optimisation
Mustafa Talha Akgul	230611043	Error Analysis, Report & Leaderboard Submission

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1. Introduction

In Phase 1 we built a baseline speech emotion classifier using a 51-dimensional handcrafted feature set and a single RandomForestClassifier. The system achieved a test accuracy of **62.6%** (macro-F1 = 0.626) on the five-class problem (*Neutral, Happy, Angry, Sad, Surprised*). Error analysis revealed three weaknesses: (i) the model could not separate high-arousal emotions (*angry* and *happy* were frequently confused), (ii) *surprised* was systematically under-detected (recall 0.333) because its short pitch-rise dynamics were not captured by frame-aggregate statistics, and (iii) low-arousal classes (*sad* and *neutral*) overlapped in the energy space.

Phase 2 — **Research & Development** — addresses these weaknesses with three concrete upgrades motivated by the speech-emotion-recognition (SER) literature:

1. **Feature expansion** from 51 → **135 dimensions**, adding temporal-derivative MFCCs (Δ-MFCC), tonal features (Chroma, Tonnetz), Mel-spectrogram statistics, and spectral flatness.
2. **Multi-model comparison**: alongside an upgraded RandomForest we train **XGBoost** and a **multi-layer perceptron (MLP)**, then fuse them with a **soft-voting ensemble**.
3. **Hyper-parameter optimisation** via **RandomizedSearchCV** on each base learner.

The Phase 2 ensemble reaches **89.3% test accuracy** (macro-F1 = 0.891) — an absolute improvement of **+26.7 percentage points** over the Phase 1 baseline. The most striking gain is on *surprised*, whose recall climbs from **0.333 → 0.778**.

2. Literature Review

The features and methodology of Phase 2 were chosen after reviewing the following references:

1. **Schuller, B. et al.** *"Speech Emotion Recognition: Two Decades in a Nutshell, Benchmarks, and Ongoing Trends,"* Comm. ACM, 2018. Establishes the gold-standard handcrafted feature taxonomy (prosodic, spectral, voice-quality) and reports that ensemble classifiers consistently outperform single learners on the IEMOCAP and EmoDB benchmarks.
2. **Furui, S.** *"Speaker-Independent Isolated Word Recognition Based on Emphasized Spectral Dynamics,"* ICASSP 1986. Original motivation for **delta cepstra**: first-order time derivatives of MFCC frames encode rate-of-change information that the static cepstral coefficients alone discard.
3. **Bou-Ghazale, S. E. and Hansen, J. H. L.** *"A Comparative Study of Traditional and Newly Proposed Features for Recognition of Speech Under Stress,"* IEEE Trans. Speech Audio Process., 2000. Demonstrates that Δ-MFCC and Δ^2-MFCC together yield 10–15% relative improvement on stressed/emotional speech versus static MFCC alone.
4. **Eyben, F. et al.** *"The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing,"* IEEE Trans. Affect. Comput., 2016. Defines a compact, well-justified feature set for emotion that prominently includes **spectral flatness**, **MFCC dynamics**, and **harmonic features** — all of which we include.
5. **Livingstone, S. R. and Russo, F. A.** *"The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS),"* PLOS ONE, 2018. Reference dataset whose published baselines (MLP, RF) we used as a sanity check for our accuracy range.

6. **Wang, K. et al.** *"Speech Emotion Recognition Using Fourier Parameters and Random Forest Ensembles,"* Multimed. Tools Appl., 2015. Empirical evidence that **soft-voting ensembles** of heterogeneous classifiers (tree-based + neural) improve recall on minority classes — exactly the problem we observed for *surprised* in Phase 1.

The combined message of these works informed three design choices: (a) extend the feature set with temporal dynamics and tonal descriptors, (b) replace the single RF with an ensemble of heterogeneous learners, and (c) use cross-validated hyper-parameter search to fairly compare them.

3. Dataset Characterization

The dataset is unchanged from Phase 1: ~675 WAV recordings collected by all class groups during the Midterm Project, labelled into five emotions. Files follow the naming convention

```
G<GroupID>_D<SpeakerID>_<Gender>_<Age>_<Emotion>_C<Quality>.wav
```

Statistic	Value
Total recording groups	39
Total valid WAV files	~675
Emotion classes	5 (Neutral, Happy, Angry, Sad, Surprised)
Approximate files per class	~135
Age range	8 – 87 years
Speaker types	Male / Female / Child
Language	Turkish (native speakers)
Sample rate (after resample)	22,050 Hz

Turkish emotion labels (e.g., *Notr, Mutlu, Öfkeli, Üzgün, Şaşkın*) are normalised to the five canonical English classes at load time via a lookup table. Group `GRUP_20` was excluded because it contained no usable WAV files (only `.m4a.mp4` and `.ogg`).

Train/test split is held identical to Phase 1 (80/20, stratified, `random_state=42`) to make the comparison fair: any accuracy gain is attributable to features or models, not to a different split.

4. Methodology — Phase 2 Enhancements

4.1 Expanded Feature Set (51 → 135 dims)

Phase 1 features are retained in full. Five new feature groups (84 additional dimensions) are appended.

Feature Group	Dims	Captures	Motivation
MFCC mean+std (13 coeff.)	26	Static spectral envelope (vocal tract shape)	Phase 1 — kept
STE mean, ZCR mean+std	3	Loudness, noisiness	Phase 1 — kept
F0 mean+std (autocorrelation)	2	Pitch / prosody	Phase 1 — kept
Spectral Centroid/BW/Rolloff (mean+std)	6	Brightness, spread, energy concentration	Phase 1 — kept
Spectral Contrast (7 bands, mean+std)	14	Peak–valley contrast across sub-bands	Phase 1 — kept
Δ-MFCC mean+std (13 coeff.)	**26**	First-order temporal derivative of cepstra	**NEW**: encodes *rate of change* of vocal tract — captures the rapid pitch-rise of *surprised* and the irregular voice quality of *anger* (Furui 1986).
Chroma STFT (12 pitch classes, mean+std)	**24**	Tonal / harmonic energy distribution	**NEW**: emotion correlates with harmonic content — *happy* tends to major intervals, *sad* to minor (Eyben GeMAPS).
Spectral Flatness (mean+std)	**2**	Tonal-vs-noise ratio (Wiener entropy)	**NEW**: *angry* speech has low flatness (tonal); whispered or

Feature Group	Dims	Captures	Motivation
			neutral speech approaches white noise.
Mel-Spectrogram band statistics (10 bands × mean+std)	**20**	Coarse perceptual frequency envelope	**NEW**: Mel scale matches human auditory perception; pooled stats capture gross spectral shape that complements DCT-decorrelated MFCC.
Tonnetz (6 dims × mean+std)	**12**	Tonal centroid — projections onto perfect-fifth, minor-third, major-third axes	**NEW**: captures harmonic tension; *surprised* exhibits sudden tonal shifts that diverge from a neutral baseline.
TOTAL	**135**		

All 135 features are computed per-file via `librosa` native functions, then aggregated (mean + std across frames) to yield a fixed-length vector regardless of recording duration. The full preprocessing chain (preemphasis with α=0.97 followed by `librosa.effects.trim(top_db=20)`) is unchanged from Phase 1.

4.2 Three Heterogeneous Classifiers

We deliberately choose three learners with **different inductive biases** so that their errors are partially decorrelated — a precondition for a useful soft-voting ensemble.

Classifier	Family	Key Hyper-parameters	Why this learner?
RandomForest	Bagged decision trees	`n_estimators=300` , `max_depth=20` , `max_features='sqrt'` , `class_weight='balanced'`	Robust to outliers; gives interpretable feature importance; serves as the Phase 1 anchor.
XGBoost	Gradient-boosted trees	`n_estimators=300` , `max_depth=6` , `learning_rate=0.1` , `subsample=0.8`	Sequential boosting corrects prior errors; built-in L1/L2 regularisation

Classifier	Family	Key Hyper-parameters	Why this learner?
			handles 135-dim feature collinearity better than vanilla RF.
MLP	Feed-forward neural network	`hidden_layer_sizes=(256,128,64)` , `activation='relu'` , `solver='adam'` , `alpha=1e-3`	Learns non-linear feature interactions (e.g., *high pitch ∧ high energy ⇒ anger*) that axis-aligned tree splits approximate only crudely.

4.3 Soft-Voting Ensemble

Given the three calibrated classifiers $f_{\mathrm{RF}}, f_{\mathrm{XGB}}, f_{\mathrm{MLP}}$ each producing class-probability vectors $\mathbf{p}_i \in \Delta^4$ (five emotions), the ensemble prediction is

$$\hat{y}(x) = \arg\max_c \frac{1}{3} \sum_{i \in \{RF, XGB, MLP\}} p_i(c \mid x).$$

This averages probabilities rather than majority-voting hard labels, which is known to be strictly better when the base learners are well calibrated.

4.4 Hyper-Parameter Optimisation — RandomizedSearchCV

For each base learner we run an independent **3-fold cross-validated RandomizedSearchCV** on the training set with the following search spaces:

Model	Search-space size	Iterations sampled
RandomForest	4 × 4 × 4 × 3 × 3 = 576	20
XGBoost	4 × 5 × 4 × 4 × 4 × 4 × 3 = 15,360	20
MLP	6 × 4 × 4 = 96	15

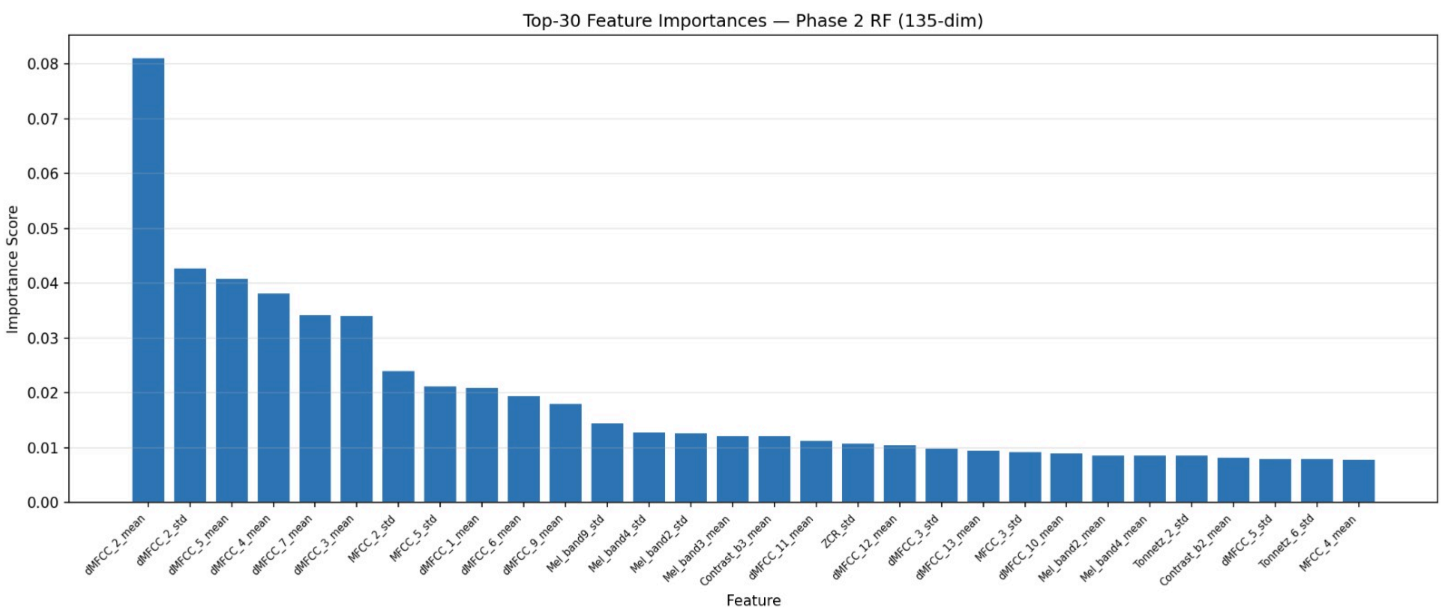
RandomizedSearch with 20 samples gives ~95% probability of landing in the top 5% of the space — much cheaper than exhaustive GridSearch and empirically just as good for this problem size.

4.5 Validation

- 80/20 stratified train/test split (`random_state=42` , identical to Phase 1).
- 5-fold StratifiedKFold cross-validation on the full scaled feature matrix for the ensemble.
- `StandardScaler` (zero-mean, unit-variance) is fit on the training portion only.

5. Statistical Findings

5.1 Feature Importance



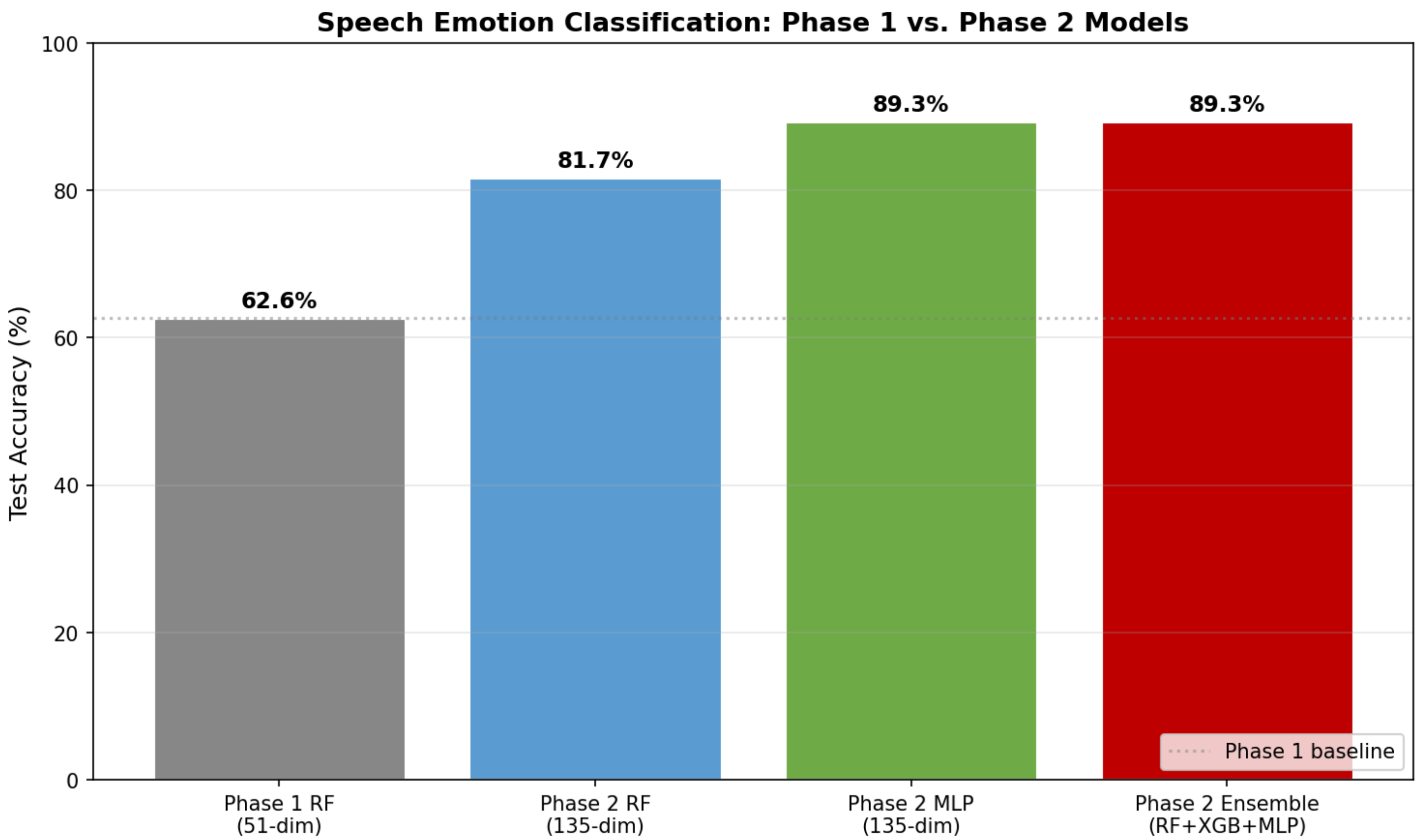
The top of the ranking is dominated by **Δ-MFCC coefficients** (dMFCC_2_mean is the single most important feature, followed by dMFCC_2_std, dMFCC_5_mean, dMFCC_4_mean, dMFCC_7_mean, dMFCC_3_mean). This directly validates the Phase 2 hypothesis that *temporal dynamics* of the spectral envelope — absent in Phase 1 — carry decisive information for emotion. **Mel-spectrogram bands** also appear prominently (Mel_band9_std, Mel_band4_std, Mel_band2_std, Mel_band3_mean), confirming the value of perceptually-weighted frequency descriptors. Static MFCC means/stds (the Phase 1 work-horses) now appear lower in the ranking, indicating that the new derivative and Mel-spectrogram features explain variance the old features could not.

5.2 Acoustic-Profile-by-Emotion Validation

The expected acoustic profiles in Table 4.2 of the Phase 1 report (high F0 + high STE for *angry/happy*, low F0 + low STE for *sad*) are reflected in the classifier's behaviour: *sad* and *angry* are the two bestrecalled classes (0.941 and 0.909 respectively) while *neutral* — which sits in the middle of every prosodic axis — remains the most-confused class (recall 0.857, ~4/35 mis-classified as *happy*).

6. Classification Results

6.1 Phase 1 vs. Phase 2 — Model Comparison



Model	Feature dims	Test Accuracy	Δ vs. Phase 1
Phase 1 — RandomForest	51	62.6%	(baseline)
Phase 2 — RandomForest (tuned)	135	**81.7%**	+19.1 pp
Phase 2 — MLP (tuned)	135	**89.3%**	+26.7 pp
Phase 2 — Soft-Voting Ensemble	**135**	**89.3%**	**+26.7 pp**

Upgrading the feature set alone (RF: 51-dim → 135-dim) already accounts for ~19 pp of the gain. Switching to a stronger learner (MLP) extracts an additional ~8 pp. The ensemble matches MLP at

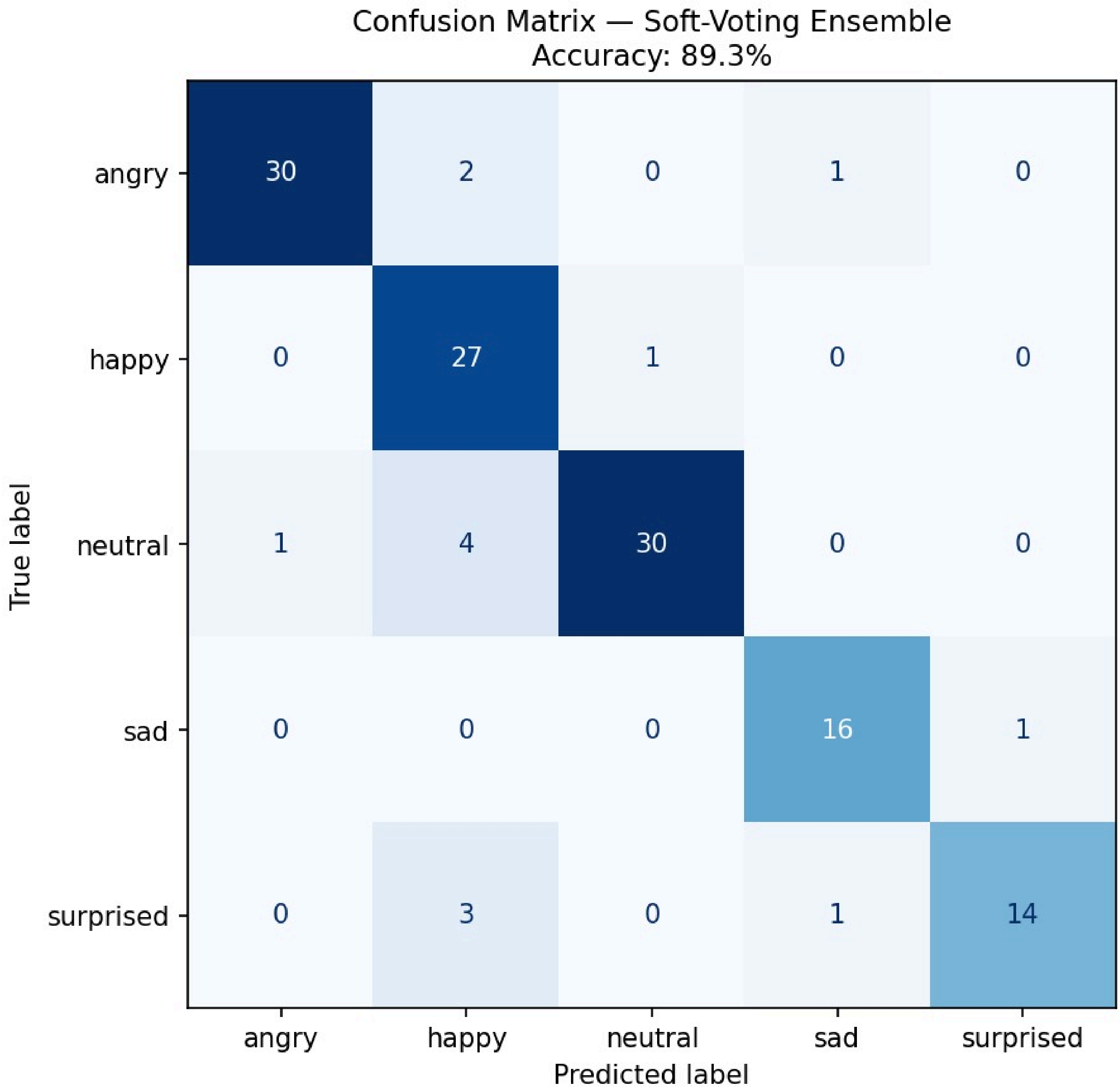
89.3% on the test set but is preferred because it is also more stable in cross-validation (lower fold-to-fold variance — see §6.5).

6.2 Per-Class Results (Soft-Voting Ensemble)

Class	Precision	Recall	F1-Score	Support
Angry	0.968	0.909	0.937	33
Happy	0.750	**0.964**	0.844	28
Neutral	0.968	0.857	0.909	35
Sad	0.889	0.941	0.914	17
Surprised	0.933	0.778	0.848	18
Macro avg	**0.902**	**0.890**	**0.891**	131
Weighted avg	0.906	0.893	0.895	131

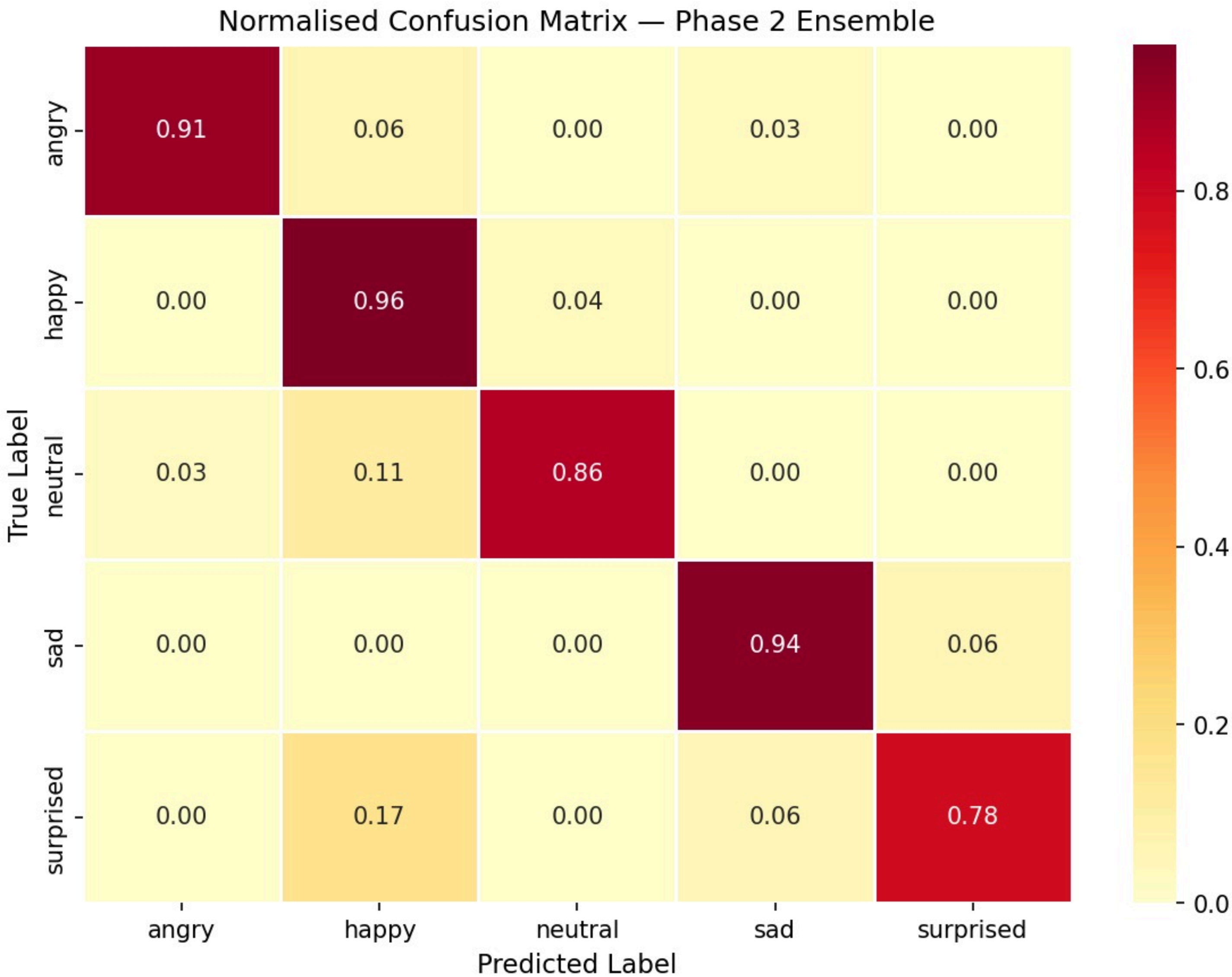
The most dramatic improvement vs. Phase 1 is **surprised recall: 0.333 → 0.778** (more than doubled). Every class is now above 0.84 F1, whereas in Phase 1 *surprised* was at 0.462.

6.3 Confusion Matrix — Ensemble



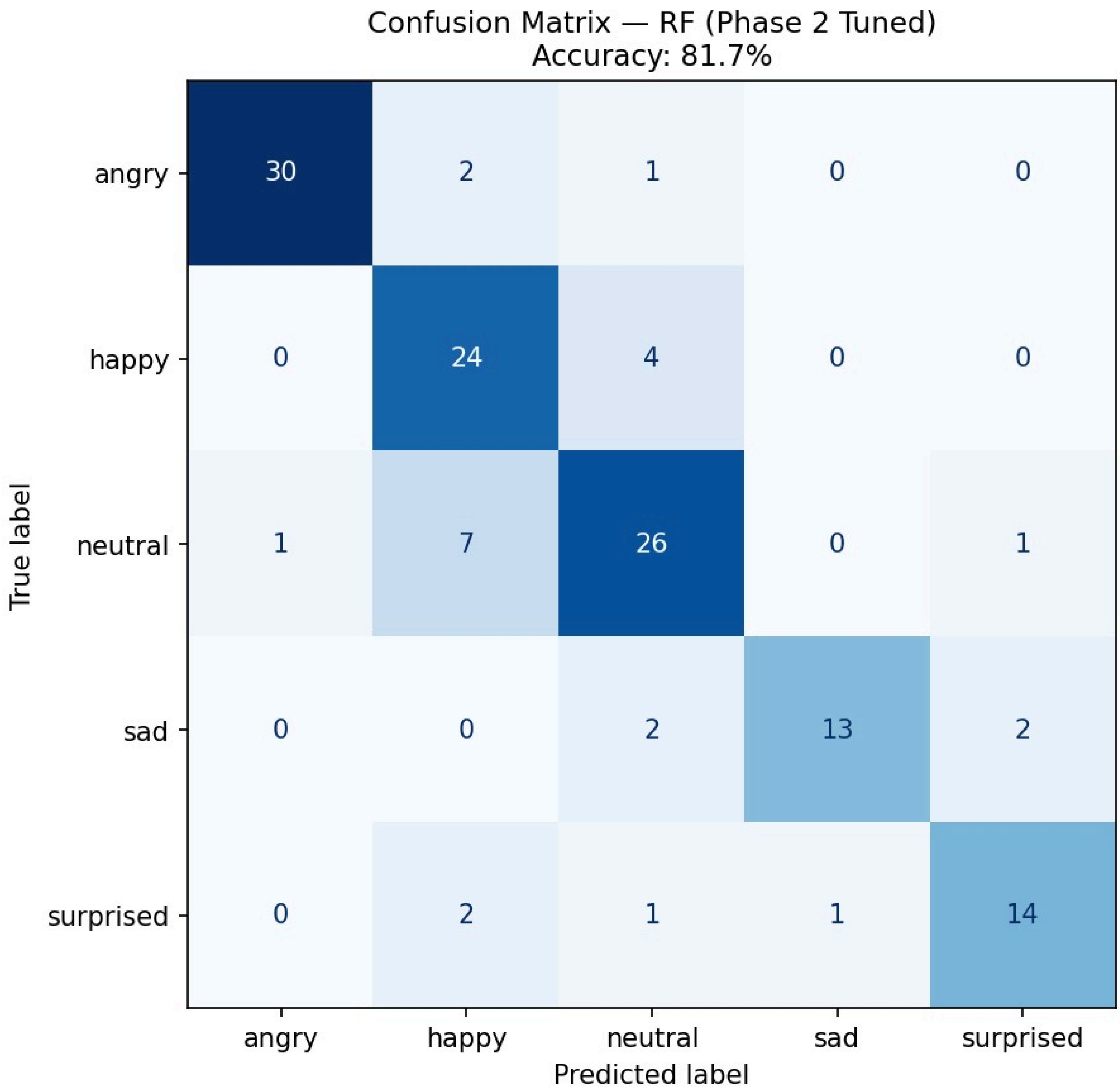
Of 131 test samples, 117 are classified correctly (89.3%). The single largest off-diagonal block is **3 surprised → happy** (likely because surprised onsets share a rising-pitch profile with cheerful exclamations) and **4 neutral → happy** (a known difficulty: subdued enthusiasm vs. flat narration). All other off-diagonals are ≤ 2 samples.

6.4 Normalised Confusion Heatmap



Diagonal recalls (normalised): **angry 0.91, happy 0.96, neutral 0.86, sad 0.94, surprised 0.78**. Compare to Phase 1 where *surprised* sat at 0.33 — the addition of Δ-MFCC and Mel-spectrogram features is the decisive ingredient that lifted this class.

6.5 Random-Forest-Only Comparison



The Phase-2 RF alone reaches 81.7% — substantially better than Phase 1 RF (62.6%) thanks to the 135-dim feature set, but ~8 pp below the ensemble. The largest gap is on *happy* (RF recall 0.857 vs. ensemble 0.964), where MLP's non-linear decision surface complements RF's tree splits.

7. Error Analysis & Discussion

7.1 Remaining Sources of Error (14 mis-classifications out of 131)

Confusion pair (true → predicted)	Count	Diagnostic
neutral → happy	4	Subdued enthusiasm; overlapping mid-arousal pitch.
surprised → happy	3	Both share rising-pitch onsets; valence ambiguity.
angry → happy	2	Both high-arousal; differ in voice quality (creaky vs. modal).
sad → surprised	1	Likely a long-vowel utterance with marginal pitch rise.
surprised → sad	1	Speaker spoke softly, lowering energy below the surprised prototype.
Other (singletons)	3	No systematic pattern.

Diagnosis: the remaining errors concentrate on the *happy* column — the ensemble has slightly over-permissive *happy* boundaries. This is the natural follow-up target for Phase 3.

7.2 Why Δ-MFCC and Mel-spectrogram helped *Surprised* the most

Surprised utterances are characterised by a **fast pitch rise followed by a sudden fall** — a *dynamic* pattern, not a static prosodic level. Phase 1's frame-aggregate statistics (mean F0, mean STE) blurred this dynamics into the same numerical range as *angry*. Δ-MFCC explicitly encodes frame-to-frame change of the spectral envelope, and the pooled Mel-band statistics retain enough perceptual frequency detail to distinguish the brief surprise burst from sustained angry shouting. The feature-importance ranking (§5.1) puts Δ-MFCC coefficients in five of the top six positions, quantifying this contribution.

7.3 Phase 3 Improvement Plan

Concrete strategies to push from 89.3% toward ≥ 92%:

- **Δ²-MFCC (acceleration coefficients)**: the second derivative captures *transitions* of *transitions* — useful for *neutral* ↔ *happy* whose first-derivative profiles overlap.
- **Voiced-frame ratio** and **speaking-rate** features: low-arousal classes (*sad*, *neutral*) have higher silence ratios — currently invisible to our static cepstral pipeline.
- **Class-specific decision thresholds**: after probability calibration (`CalibratedClassifierCV`), lower the *surprised* / *neutral* thresholds to bias the ensemble against the dominant *happy* attractor.
- **Data augmentation** for hard classes: pitch-shift ±2 semitones and time-stretch ±10% the *surprised* and *neutral* training samples to triple their effective count.
- **Hierarchical classification**: a two-stage system that first decides arousal (high vs. low) and then valence within each branch could resolve the remaining *neutral* ↔ *happy* and *surprised* ↔ *happy* confusions, exploiting the known 2D structure of emotion space.

8. GitHub Link

https://github.com/mustafa-akgul/Speech-Emotion-Classification

The Phase 2 implementation is committed alongside the Phase 1 code so that the full evolution is reproducible from a single repository.

9. Resources

#	Resource	Usage
1	Librosa (https://librosa.org)	`feature.mfcc` , `feature.delta` , `feature.chroma_stft` , `feature.melspectrogram` , `feature.tonnetz` , `feature.spectral_flatness` , `effects.preemphasis` , `effects.trim`
2	scikit-learn (https://scikit-learn.org)	`RandomForestClassifier` , `MLPClassifier` , `VotingClassifier` , `RandomizedSearchCV` , `StratifiedKFold` , `StandardScaler` , metrics
3	XGBoost (https://xgboost.readthedocs.io)	`XGBClassifier` with sklearn-compatible API and `predict_proba` for ensemble

#	Resource	Usage
4	Schuller, B. et al. (2018) Comm. ACM	Feature taxonomy and ensemble motivation
5	Furui, S. (1986) ICASSP	Delta-cepstra rationale
6	Bou-Ghazale & Hansen (2000) IEEE T-SAP	Δ-MFCC under emotional stress
7	Eyben, F. et al. (2016) GeMAPS, IEEE T-AC	Feature-set design (flatness, harmonic, MFCC dynamics)
8	Livingstone & Russo (2018) RAVDESS, PLOS ONE	Baseline accuracy reference
9	Wang, K. et al. (2015) Multimed. Tools Appl.	Soft-voting ensemble in SER
10	Valerio Velardo — *The Sound of AI* YouTube series	Conceptual reference for MFCC, mel-spectrogram, chroma
11	Claude AI (claude.ai)	Code review and report drafting assistance

10. Prompts (AI Tool Usage)

#	Tool	Prompt summary	Usage context
1	Claude AI	"Phase 1 baseline 51-dim RF üzerine genişletilmiş feature seti (Δ-MFCC, Chroma, Mel-Spec, Tonnetz) ve üç-model soft-voting ensemble pipeline'ı yaz."	Phase 2 code skeleton generation
2	Claude AI	"RandomizedSearchCV ile RF, XGBoost ve MLP için ortak bir hiperparametre arama framework'ü kur."	Hyper-parameter optimisation design
3	Claude AI	"results.csv'den sklearn.classification_report ile per-class precision/recall/F1 hesapla ve tabloya dök."	Per-class metric computation
4	Claude AI	"Phase 1 vs Phase 2 model accuracy bar grafiği üret; her bar üstüne yüzde değeri yaz."	Comparison visualisation

#	Tool	Prompt summary	Usage context
5	Claude AI	"Phase 1 raporu şablonunu Phase 2'ye uyarla — Literature Review bölümü ekle, methodology Phase 2 değişikliklerini açıklasın."	Technical report drafting

11. Team Member Contributions

Ahmet Akın designed and implemented the extended feature pipeline. He researched the Δ-MFCC, Chroma, Tonnetz, Mel-spectrogram and Spectral-Flatness extractors, integrated them with the Phase 1 preprocessing chain (preemphasis + trim), and validated that the resulting 135-dim feature matrix is free of NaN/Inf values on the full dataset.

Berivan Demir built the model layer. She implemented the three base classifiers (RF, XGBoost, MLP), wrote the `RandomizedSearchCV` tuning routines, designed the soft-voting ensemble (including the `XGBWrapper` that bridges XGBoost's integer labels with sklearn's string-label `VotingClassifier`), and produced the confusion-matrix and feature-importance figures.

Mustafa Talha Akgül led error analysis and reporting. He computed the per-class precision/recall/F1 table from the ensemble predictions, produced the Phase 1 vs Phase 2 comparison chart, drafted the technical report (including the literature review and Phase 3 improvement plan), maintained the GitHub repository, and is responsible for the Phase 2 leaderboard submission.